ISHAN MALAWIGE

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About Me

Dedicated ICT undergraduate with strong interest in computer networking and network security. Skilled in configuring and troubleshooting routers, switches, firewalls, VLANs, NAT, DNS, and DHCP, supported by multiple Cisco certifications. Possesses a solid understanding of ICT infrastructure and cybersecurity fundamentals, with a goal of building secure, efficient, and scalable network systems. Adaptable, quick to learn, and motivated to grow in the networking and cybersecurity field.

EDUCATION

1. General Sir John Kotelawala Defence University-Ratmalana Sri Lanka
   BT(Hons)Information and Communication Technology 2023-present

2. President's Central College - Embilipitiya
   GCE A/L (Technology) 2021 2013-2021

SKILLS SUMMARY

- **Networking:** TCP/IP, DNS, DHCP, OSPF,RIP, VLANs, NAT, Port Security
- **Network Devices:** Routers, Switches, Firewalls(FortiGate. Sophos)
- **Tools & Platforms:** Wireshark, Packet Tracer, EVE-NG
- **Cloud & Virtualization:** VMware ESXI ,vSphere, Hyper-V
- **Systems:** Windows Server, Active Directory (User & Group Management, GPO), Linux (Basic)
- **Soft Skills:** Effective communication, Teamwork & collaboration, Adaptability & flexibility, Time management & organization, Quick learning & continuous improvement, Problem-solving & analytical thinking

EXPERIENCE

Network Engineer-Intern (6 months)

Sri Lanka Telecom PLC– Sri Lanka | July 2025 –Dec 2025

- Providing Level 1 support for company-wide network systems and users.
- Keeping updated records of IT assets, configurations, and issue resolutions.
- Configuring and maintaining routers, switches, and MSANs
- Assisting the Group IT team with network security policy enforcement, and infrastructure upgrades.
- Monitoring network performance and responding to incidents related to connectivity and latency.

Demonstrator (2026 July-Present)

Vocational Training Center, Dehiwala

- **Microsoft 365 Administration:** manage the M365 Admin Center, handling user lifecycle management including account creation, offboarding, password resets
- **Technical Support & Lab Facilitation:** Provide real-time technical support during hands-on lab practicals, troubleshooting student access, system, and network
- **User & Access Management:** Implement access management policies to maintain security and seamless system entry for students

Projects

Redesign and Optimization of an Existing University Network Infrastructure (Simulation)
General Sir John Kotelawala Defence University | 2026 (ongoing)

- Designed a scalable enterprise network using layered architecture (core, distribution, access) with VLAN segmentation and secure routing.
- Implemented dynamic routing using OSPF and configured inter-VLAN communication to optimize traffic flow and network efficiency.
- Configured FortiGate firewall policies with High Availability (HA) to ensure continuous network security and failover protection.
- Integrated LAN and wireless network infrastructure into a unified architecture for consistent management and policy enforcement.
- Proposed and designed multi-ISP connectivity with redundant network paths to enhance availability and minimize service disruption.
- Implemented server virtualization using VMware vSphere and ESXi to improve resource utilization and scalability.
- Designed and integrated dedicated NAS and SAN storage solutions to support virtual machines and ensure high-performance data access.
- Developed a Disaster Recovery (DR) strategy with backup and failover mechanisms to ensure business continuity.
- Simulated and tested network performance, redundancy, and failover scenarios using a virtual lab environment (EVE-NG).

Technologies: VLAN, OSPF, EVE-NG, FortiGate Firewall (HA), VMware ESXi, vSphere, NAS, SAN, Active Directory

Data Center and Network Administration Project (Simulation)

Vocational Training Center | 2026

- Designed and implemented a domain-based enterprise network infrastructure for a 350+ employee organization using Windows Server 2025.
- Installed and configured Active Directory Domain Services (AD DS), Organizational Units (OUs), departmental user accounts, and Group Policy for centralized security and management.
- Configured security policies including password complexity, password expiration, account lockout, screen lock, and department-based restrictions.
- Configured DHCP Server with IP address scopes, reserved address ranges, and DHCP Failover for network availability.
- Implemented File Server Resource Manager (FSRM) with departmental storage quotas, file screening, and folder permissions.
- Configured IIS to host a company website and Windows Deployment Services (WDS) for Windows operating system deployment.
- Configured RSAT to provide administrative capabilities for the Finance Department.
- Deployed Windows Server 2025 on Microsoft Azure as an Additional Domain Controller to improve Active Directory availability.

Technologies: Windows Server 2025, Active Directory, Group Policy, DHCP, FSRM, IIS, WDS, Microsoft Azure, RSAT.

Enterprise Network Modernization: Centralized AD/DHCP & Multi-Site FortiGate Security

Vocational Training Center | Sep 2026

- **Centralized Identity & Infrastructure:** Configured cross-site DHCP Relay agents on FortiGate firewalls to deliver dynamic IP leasing from a central AD/DHCP server (192.168.100.2) across Core VLANs and remote branch subnets (192.168.120.0/24), successfully domain-joining all enterprise endpoints (IT, HR, Branch).
- **Multi-Site Routing & Segmentation:** Designed and deployed end-to-end static routing across an ISP core and established stateful, non-NAT FortiGate firewall policies enforcing zero-trust, bidirectional segmentation between Core, Server, and Branch zones.
- **Granular RBAC & Web Filtering:** Provisioned custom, role-based FortiGuard web filtering profiles (unrestricted for IT; targeted access for standard/privileged HR staff) and administrative RBAC tiers (Full Admin vs. Read-Only Auditor access).

Technologies: Active Directory ,VLAN ,PNET Lab , DHCP

CERTIFICATES

Cisco Certifications:

- CCNA: Introduction to Networks
- CCNA: Switching, Routing, and Wireless Essentials
- CCNA: Enterprise Networking, Security, and Automation
- Cisco Network Security
- Cisco Ethical Hacker

Research Publications

• M. I. Sathsara, A. Kurunayaka, & R. Laveena (2024). *A Comprehensive Framework for Strengthening IoT Security: Mitigating Vulnerabilities through Advanced Mechanisms in Industrial Systems.* International Research Conference, General Sir John Kotelawala Defence University. Paper ID: 587. Available at: https://irc.kdu.ac.lk/2024/publications/files/fot.pdf

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References

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I hereby declare that the information provided above is true and accurate to the best of my knowledge and belief.



Ishan Malawige
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